Advanced Wireless Communications Laboratory Series

Closed-Loop Cognitive Spectrum Access with Adaptive Sensing

GNU Radio and Python simulation laboratory

Reproducible code, recorded measurements, and analysis

WC-LAB-03 2026

Executive summary

This laboratory closes the loop between wideband spectrum sensing and secondary channel access. A 128-point FFT resolves eight simulated frequency channels. An energy detector tracks the receiver noise floor, while a causal controller keeps, vacates, or changes its selected channel in response to detected primary activity. Sensing mutes secondary transmission; switching channels adds a retuning guard. The study therefore evaluates sensing decisions and their consequences for spectrum access in the same experiment.

Three policies are compared on matched primary-traffic and noise realizations: a fixed threshold and sensing interval; an adaptive noise reference with a fixed interval; and the same adaptive reference with an adaptive interval. The dataset contains 27 runs across three traffic conditions, three policies, and three seeds. Each run provides 5.92 s of evaluated access after an 80-ms calibration period.

In nominal traffic, adaptive-period sensing reduces mean primary/secondary overlap from 321.3 to 174.7 ms per run relative to fixed-period sensing with the same detector, a 45.6% reduction. Successful-payload goodput falls from 56.01 to 52.55 kb/s because sensing uses transmission time. These are results from a specified collision-erasure packet model, not physical-layer packet measurements or measured hardware retuning delays.

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1 Introduction

A cognitive spectrum-access receiver must do more than recognize an occupied channel. Its sensing decisions must control secondary transmission quickly enough to limit interference with returning primary users, while leaving sufficient time for useful communication. This sensing–access feedback loop is a defining feature of cognitive radio [1].

Energy detection depends on an estimate of receiver noise power. If the noise floor increases while the threshold remains fixed, vacant channels may be reported occupied. Noise uncertainty also limits the detection of weak signals [2]. The experiment tests both effects using an explicit noise drift and a separate weak-primary condition.

Wideband spectrum sensing is also the subject of sparse-transform approaches, including the work of Hassanieh et al. [3]. This laboratory instead uses a conventional full-rate FFT so that the detector, access schedule, and resulting transmission outcomes can be evaluated together. The central comparison is between accurate but infrequent sensing and more frequent sensing that consumes additional transmission opportunities.

2 System architecture

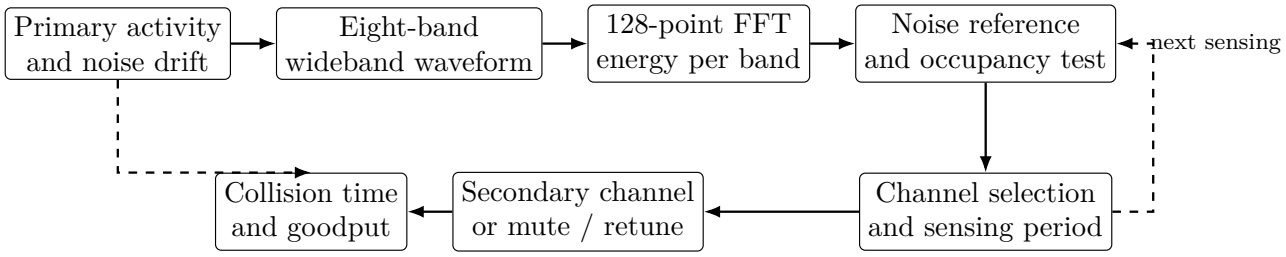


Figure 1: Closed-loop simulation. The detector receives primary-signal observations only on scheduled sensing intervals. Primary occupancy is known to the simulation for scoring, not to the controller.

The sensor is silent with respect to the secondary transmitter during a sensing interval; this avoids assuming perfect self-interference cancellation. Primary transmissions are exogenous and do not respond to the secondary user. The controller changes the selected channel only after receiving a sensing result. No future primary state or true noise level is provided to it.

2.1 Wideband signal and traffic

The simulation has $C = 8$ channels, each occupying $B = 16$ contiguous bins of a unitary $N = 128$ point FFT. Eight independent 128-sample observations are taken in each sensing interval. At the modeled sample rate $F_s = 512$ ksample/s, the resulting $MN = 1024$ samples occupy $T_s = 2$ ms. Every channel therefore spans 64 kHz in the frequency-domain grid.

Let $u_c[t] \in \{0, 1\}$ indicate whether primary channel c is active in discrete interval t. The primary state follows an independent two-state Markov model with

$$\Pr(u_c[t] = 1 \mid u_c[t-1] = 0) = p_{01}, \qquad \Pr(u_c[t] = 0 \mid u_c[t-1] = 1) = p_{10}. \tag{1}$$

Active primary bins carry independent unit-energy QPSK symbols with fixed absolute received power. A unitary inverse FFT forms the time-domain wideband signal; complex Gaussian receiver noise is added; the receiver FFT recovers the channelized observations. Receiver noise power increases from 0 to 4 dB

over each run, with a small deterministic variation of 0.25 dB. Thus the primary-to-noise ratio falls as the noise floor rises.

The simulated primary bands are aligned with FFT channel boundaries. No adjacent-channel leakage, frequency offsets, shadowing, or RF front-end nonlinearity is modeled.

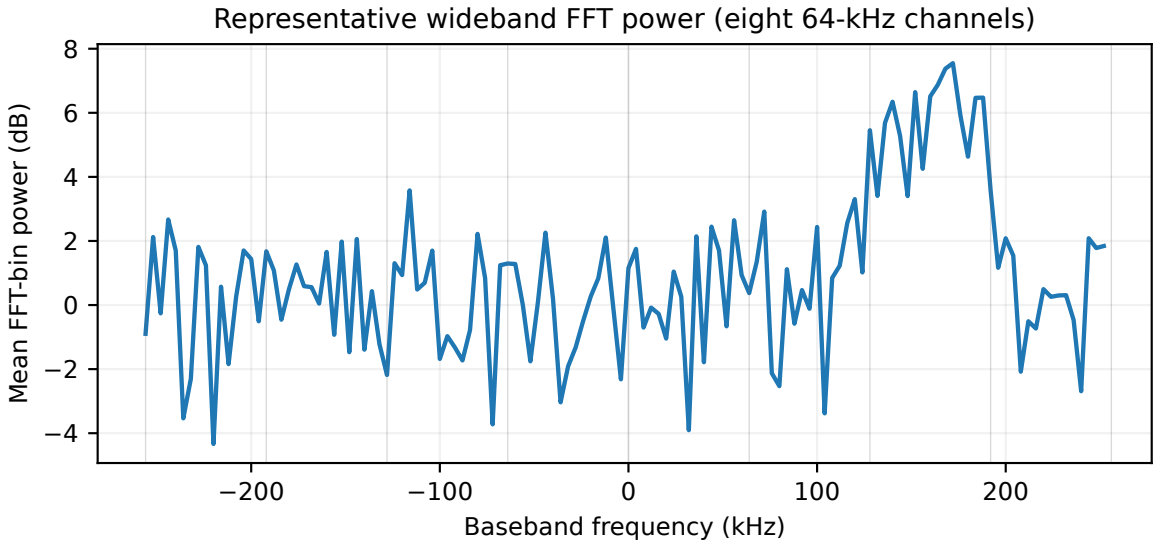


Figure 2: One recorded wideband FFT power estimate. The vertical guides separate 64-kHz sub-bands; the displayed response is an example, not an occupancy estimator averaged over the full experiment.

3 Adaptive sensing and access control

3.1 Energy detector and noise reference

For the set $\mathcal{B}_c$ of 16 bins assigned to channel c, the test statistic at a sensing interval is

$$E_c[t] = \frac{1}{MB} \sum_{m=0}^{M-1} \sum_{k \in \mathcal{B}_c} |Y_{t,m}[k]|^2, \qquad M = 8,\ B = 16. \tag{2}$$

With complex Gaussian noise alone, $\mathbb{E}[E_c] = \sigma_w^2$. Before scored operation, the receiver observes 40 explicitly vacant 2-ms intervals and computes an initial noise reference $\widehat{\sigma}_0^2$. This 80-ms calibration period is excluded from performance statistics.

For subsequent sensing instances, let $L_3[t]$ be the mean of the three smallest channel energies. A fixed calibration factor b is determined from the initial noise-only measurements to compensate for downward order-statistic bias. The reference is updated only when sensing occurs:

$$\widehat{\sigma}_t^2 = 0.60\widehat{\sigma}_{t^-}^2 + 0.40\, bL_3[t], \qquad \widehat{u}_c[t] = \mathbb{1}\left\{E_c[t] > 1.20\widehat{\sigma}_t^2\right\}. \tag{3}$$

The alternative fixed-threshold policy retains $1.20\widehat{\sigma}_0^2$. The multiplier 1.20 is an experimentally specified setting, not an assertion of a guaranteed constant false-alarm probability. Estimation is less reliable when most channels are occupied.

3.2 Causal channel selection and retuning

After sensing, the controller keeps its current channel if the detector still classifies that channel as free. If that channel is declared occupied, it selects the lowest-energy channel among those classified as

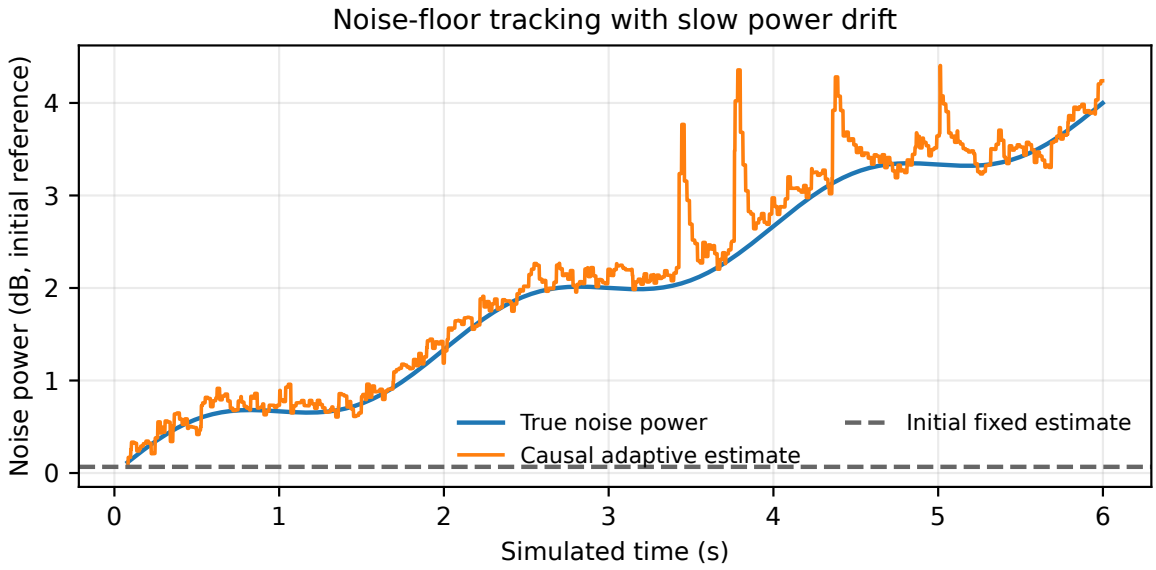


Figure 3: Noise-reference update in the nominal adaptive run (seed 2026). The true noise curve is included for evaluation only. The controller has access to its own estimate, not that curve.

vacant. When no channel is available, it stops transmitting until a later sensing event. The selection uses estimated energies only, not true occupancy.

Each sensing event mutes secondary transmission for 2 ms. When a new channel is selected, the next 2-ms interval is reserved for retuning; transmission resumes two ticks after the sensing interval began. The resulting *modeled sensing-start-to-retune-completion delay* is 4 ms for a completed switch. It is a schedule parameter, not measured radio-hardware latency.

The fixed-period policies sense every 16 ticks (32 ms). The adaptive-period policy schedules its next sensing event after 2, 4, or 8 ticks, depending on whether a switch or idle state occurred and how close current-channel energy is to the decision threshold:

$$\Delta_t = \begin{cases} 2, & \text{switch or idle, or } E_c/\lambda \geq 1, \\ 4, & 0.90 \leq E_c/\lambda < 1, \\ 8, & E_c/\lambda < 0.90. \end{cases} \tag{4}$$

All control decisions use the latest available sensing result. Traffic can change between sensing opportunities, creating a nonzero risk of overlap.

4 Performance measures

Detection quality is measured only at the controller's actual sensing times. Using the simulator's primary-occupancy truth for scoring,

$$\widehat{P}_d = \frac{N_{\text{detected busy}}}{N_{\text{sensed busy}}}, \qquad \widehat{P}_{fa} = \frac{N_{\text{declared busy}|\text{vacant}}}{N_{\text{sensed vacant}}}. \tag{5}$$

These are conditional proportions over channel-sensing observations. Because policies sense at different times, the class counts differ between policies. Successive primary states are correlated; the denominators should not be interpreted as independent Bernoulli trials.

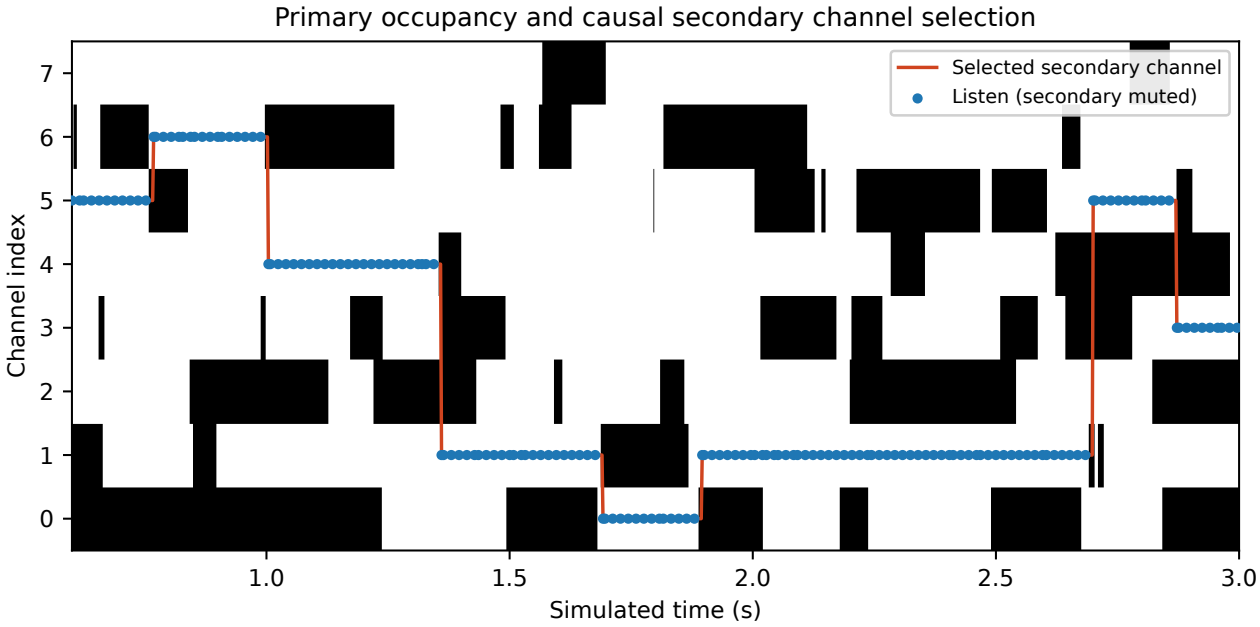


Figure 4: Primary activity (dark cells) and secondary channel selection in a recorded nominal run. Red marks the selected secondary channel; blue sensing markers indicate intervals when the secondary transmitter is muted.

A logical secondary packet contains $L = 128$ payload bits and occupies one eligible 2-ms interval. A packet is marked successful if the secondary transmits and the primary is absent on the selected channel; any overlap erases the packet. The reported secondary goodput is

$$G = \frac{LN_{\text{successful}}}{T_{\text{eval}}}, \qquad T_{\text{collision}} = T_s \sum_t \mathbb{1}\{\text{TX}[t] \wedge u_{c[t]}[t]\}. \tag{6}$$

Goodput includes sensing, idle, retuning, and collision losses. Its theoretical upper bound under this packet model is 64 kb/s. No secondary packet decoder, packet error correction, or SINR-to-packet-error relationship is simulated.

For primary returns that occur while the secondary is actively transmitting, the onset-to-vacate delay is measured until the secondary leaves that channel. A return that ends before the secondary vacates is excluded from the median vacate-delay statistic. Silent sensing gaps alone do not count as channel vacation. Cumulative collision time, by contrast, includes all primary/secondary overlap intervals, including missed detections of already occupied channels.

5 Recorded experiment

The nominal and weak-primary cases use $(p_{01}, p_{10}) = (0.008, 0.015)$ per tick. The busy-band case uses $(0.016, 0.015)$. Initial primary-to-noise ratios are $+5$ dB, -4 dB, and $+5$ dB, respectively. A given scenario and seed use the same traffic, noise, and wideband observation sequence for every policy. The measurements are computed from the simulated traffic and receiver observations. Results from three seeds are reported for each policy.

Table 1: Simulation configuration. Each scenario is evaluated on three seeds (2026–2028), with identical primary and receiver-noise samples replayed to every access policy.

Parameter	Value	Parameter	Value
FFT length	128	Channels	8
Bins / channel	16	FFT snapshots / tick	8
Time per tick	2 ms	Sample rate	512 ksample/s
Calibration	40 ticks (80 ms)	Evaluation	2960 ticks (5.92 s)
Threshold multiplier	1.20	Noise drift	0 to +4 dB
Adaptive intervals	4, 8, 16 ms	Fixed interval	32 ms
Payload	128 bits / tick	Retune guard	2 ms

6 Results

6.1 Detection and false alarms

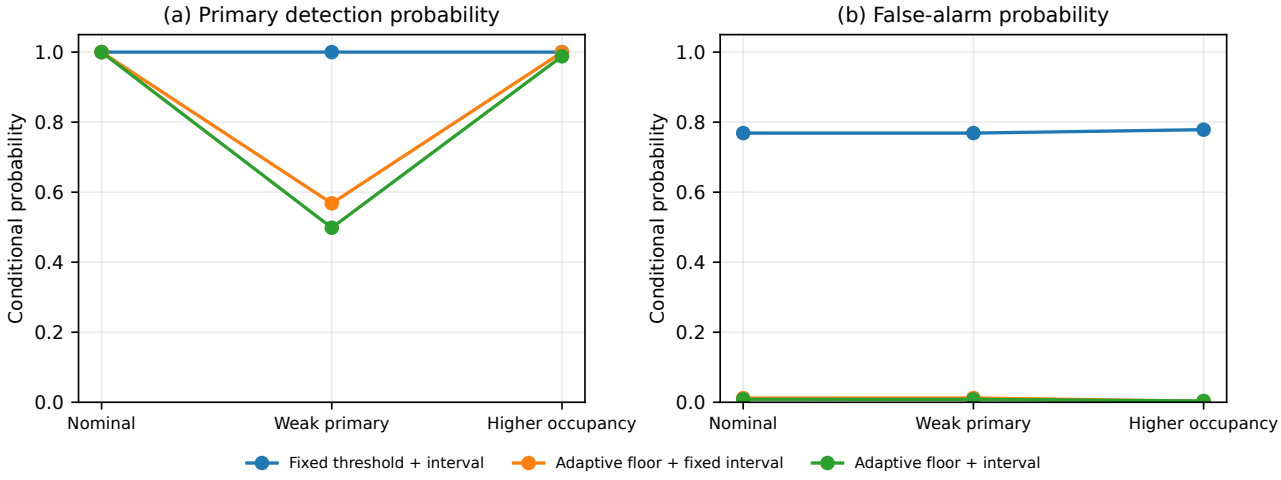


Figure 5: Mean observed detection and false-alarm proportions across three seeds. The weak-primary case exposes missed detections; a fixed absolute threshold becomes overly conservative as noise power rises.

In the fixed-threshold case, the rising noise level eventually causes many vacant channels to be declared occupied. The measured high detection fraction therefore does not represent a useful sensing solution: it coincides with a false-alarm fraction near 0.77. The adaptive noise reference markedly reduces that false-alarm fraction. Under the weaker primary, detection remains difficult because the primary signal increasingly approaches the noise floor. The slightly different detection estimates for the two adaptive-threshold policies are conditional on different sensing schedules, not evidence of a better single-window detector.

Table 2: Mean detection probabilities, false-alarm probabilities, and sensing duty fraction (three seeds per cell).

Scenario	Controller	$\widehat{P}_d$	$\widehat{P}_{fa}$	Sensing (%)
Nominal	Fixed threshold / period	1.000	0.769	6.25
	Adaptive floor / fixed period	1.000	0.012	6.25
	Adaptive floor / period	1.000	0.008	14.0
Weak primary	Fixed threshold / period	1.000	0.769	6.25
	Adaptive floor / fixed period	0.568	0.012	6.25
	Adaptive floor / period	0.498	0.008	14.1
Busy band	Fixed threshold / period	1.000	0.779	6.25
	Adaptive floor / fixed period	1.000	0.003	6.25
	Adaptive floor / period	0.988	0.004	14.0

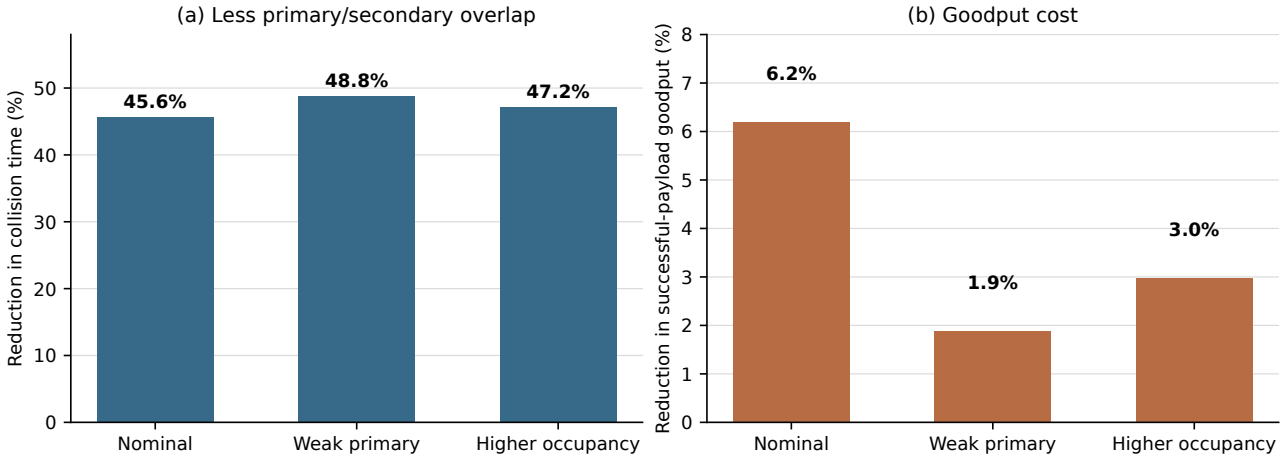


Figure 6: Change caused by replacing fixed-period sensing with adaptive-period sensing while keeping the adaptive noise reference unchanged. Percentages use the fixed-period result as the denominator; bars are computed from the means of three matched runs. Lower collision time and lower goodput occur together.

6.2 Access reliability and throughput

Table 3: Mean secondary goodput and cumulative collision time (three seeds).

Scenario	Controller	Goodput (kb/s)	Collision time (ms)
Nominal	Fixed threshold / period	18.70	89.3
	Adaptive floor / fixed period	56.01	321.3
	Adaptive floor / period	52.55	174.7
Weak primary	Fixed threshold / period	18.70	89.3
	Adaptive floor / fixed period	51.21	772.7
	Adaptive floor / period	50.26	395.3
Busy band	Fixed threshold / period	15.78	236.0
	Adaptive floor / fixed period	51.93	645.3
	Adaptive floor / period	50.39	340.7

With the adaptive noise reference held constant, shortening the sensing period reduces mean collision time by 45.6% in nominal traffic, 48.8% with weak primary signals, and 47.2% under higher occupancy. Goodput decreases by 6.2%, 1.9%, and 3.0%, respectively. More frequent sensing permits earlier channel vacation but consumes time that would otherwise carry payload.

The fixed-threshold policy has the shortest nominal overlap largely because noise-driven false alarms suppress secondary transmissions. Its nominal goodput is only 18.70 kb/s. Thus collision time cannot be interpreted without considering goodput and false alarms. The adaptive reference improves access under noise drift; it does not guarantee detection of weak primary signals.

6.3 Event timing

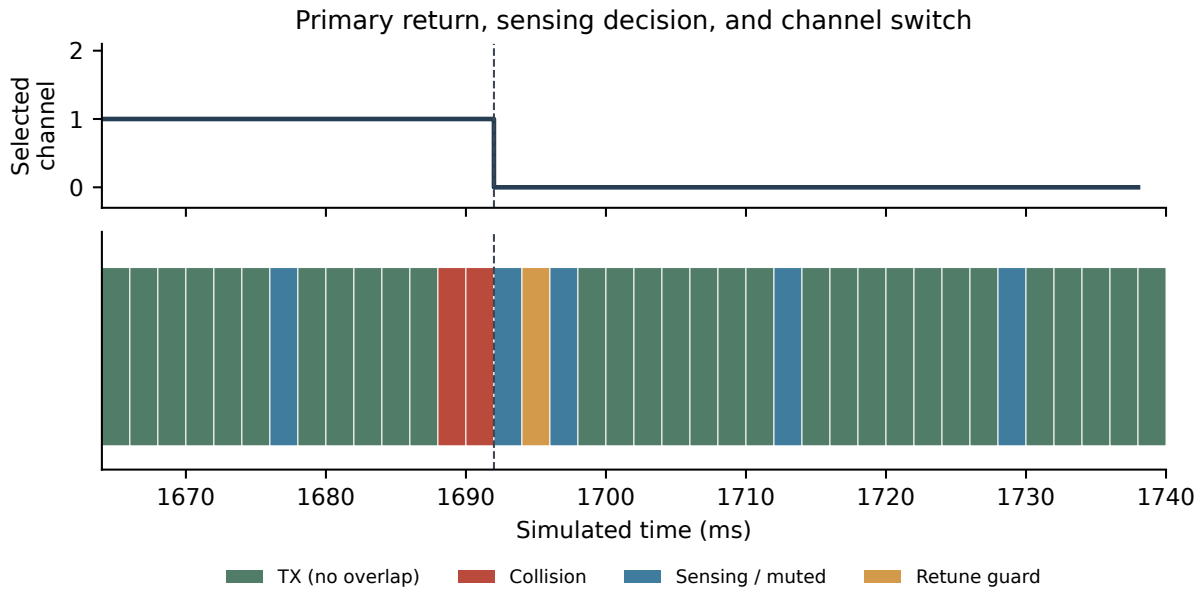


Figure 7: A recorded primary return followed by channel vacation in the nominal adaptive run (seed 2026). Each block represents one 2-ms interval; the retune guard follows the sensing decision at 1692 ms.

A completed channel switch has a modeled 4-ms sensing-start-to-resumption delay, comprising one 2-ms sensing interval and one 2-ms retune guard. In nominal traffic, the median primary-onset-to-vacate time, averaged over three seeds, is 13.0 ms for fixed-period sensing and 6.7 ms for adaptive-period sensing. These delays arise from schedule and traffic timing, not hardware bench measurements. The event-based median excludes returns that disappear before channel vacation.

7 Implementation and reproducibility

The Python simulation generates primary traffic and received signals. It performs FFT channelization, updates the noise reference, and controls secondary access. All three policies use the same traffic and noise realizations. The CSV contains 27 runs. Control traces and a representative spectrum capture are included.

The GNU Radio Companion flowgraph displays the spectrum and controller state. Automated tests check detection, calibration, sensing and retuning, and goodput calculations.

8 Assumptions and limitations

The primary signals are synthetic, FFT-bin-aligned QPSK waveforms. The model omits fading, adjacent-channel leakage, carrier offsets, propagation delay, front-end nonlinearities, and spatial hidden terminals. Secondary sensing is half-duplex. Packet success follows a collision-erasure rule rather than SINR-dependent decoding; retuning time is a specified guard, not a hardware measurement.

The noise-reference rule has no guaranteed false-alarm probability under arbitrary occupancy or noise statistics. Its low-energy-band estimate may be biased when many primary channels are active. The measurements compare the three sensing policies under the same simulated traffic conditions.

9 Conclusions

The experiment shows that spectrum access depends on both sensing accuracy and the time at which a sensing result is acted upon. A fixed energy threshold produces excessive false alarms as receiver noise rises. Updating the noise reference restores useful secondary transmission opportunities, although weak primary signals remain difficult to detect.

For nominal traffic, adapting the sensing period reduces cumulative primary/ secondary overlap by 45.6% relative to fixed-period sensing with the same noise-reference estimator, at a 6.2% goodput cost. The result quantifies the tradeoff between primary-user collision time and secondary goodput under the specified traffic conditions.

References

[1] S. Haykin, “Cognitive radio: Brain-empowered wireless communications,” *IEEE Journal on Selected Areas in Communications*, vol. 23, no. 2, pp. 201–220, Feb. 2005. doi:10.1109/JSAC.2004.839380.

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[3] H. Hassanieh, L. Shi, O. Abari, E. Hamed, and D. Katabi, “GHz-wide sensing and decoding using the sparse Fourier transform,” *Proc. IEEE INFOCOM*, pp. 2256–2264, 2014. doi:10.1109/INFOCOM.2014.6848169.